25901189 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCSC200,MG4DSCECC201 - OOPS CONCEPTS USING JAVA

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Illustrate the use of arithmetic and relational operators with a simple Java example. [A] / [CO1]
2. Explain how a base class constructor is accessed from a derived class constructor in Java and mention the keyword used for this purpose. [U] / [CO2]
3. Name the Swing class used to create a tick box in Java. [K] / [CO3]
4. Identify and Examine the purpose of an interface in Java. [An] / [CO2]
5. State any two characteristics of a constructor. [K] / [CO2]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. What are labelled loops in Java? Explain their purpose with syntax. [U] / [CO1]

7. Illustrate various ways of initializing one-dimensional arrays and examine their differences. [An] / [CO2]

8. Use the Delegation Event Model to explain how events are handled in Java. [A] / [CO3]

9. Explain Polymorphism. [U] / [CO1]

10. Describe abstract classes and abstract methods in Java. [K] / [CO2]

11. Explain the role of javax.swing and java.applet packages in Java applications. [K] / [CO3]

12. Identify the different types of inheritance supported by Java and illustrate each with suitable examples. [U] / [CO2]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Discuss the key features of Java in detail. [K] / [CO1]

14. Explain how two methods can be defined in a class and how can the user calls it from main program? Illustrate with an example. [U] / [CO2]

[15x1 = 15]